Sanad User Stories

1. Stakeholder Categories and System Roles

This section introduces the people and organizations involved in Sanad and explains how they take part in the service.

Stakeholder category	**Identified roles**	**Role in the Sanad project**	**Stories**
1. Internal Operational (Daily workflow staff)	Care Coordinator Caregiver Team Lead Field Caregiver	Coordinates member setup, consent and routines; schedules caregivers; records and verifies visits.	US-01 to US-04
2. Internal Executive (Management)	Founder and Director	Oversees service quality, subscriptions, invoices, caregiver records and payroll.	US-05 to US-07
3. External Operational (Service recipients, family and community)	Elderly Member Family Contact Neighbor Volunteer	The member uses the smart speaker and privacy controls; family contacts receive summaries and alerts, send voice messages, book and rate visits, and subscribe; volunteers respond to non-urgent alerts.	US-08 to US-17
4. External Executive (External service stakeholders)	Technology Partner Emergency Services (123)	The technology partner supports device reliability; emergency services receive hard-fall dispatch information.	US-18, US-19

2. User Stories

The nineteen stories below describe what each stakeholder needs from Sanad and why it matters. Each follows the standard "As a [role], I want to [goal], so that [benefit]" format and describes a need, not a technical solution.

US-01 Care Coordinator | Internal Operational

As a Care Coordinator, I want to set up and update each member's daily routine, including wake-up time, meal times, and medication reminders, so that the system uses the member's normal routine when it monitors for unusual activity.

US-02 Care Coordinator | Internal Operational

As a Care Coordinator, I want to record each member's informed consent and complete the six-month consent reviews, so that monitoring only takes place with valid consent.

US-03 Caregiver Team Lead | Internal Operational

As a Caregiver Team Lead, I want to assign caregivers to requested visits based on district, skills, language, availability, and the ten-hour daily working-hours limit, so that members receive suitable care and no caregiver is overworked.

US-04 Field Caregiver | Internal Operational

As a Field Caregiver, I want to confirm my arrival and departure with check-in and check-out codes and submit a short visit note, so that completed visits are properly verified and recorded.

US-05 Founder and Director | Internal Executive

As the Founder and Director, I want to review service information such as the number of false alerts, average alert response time, how alerts were resolved, caregiver family ratings, and members' activity trends, so that I can assess service quality and make informed decisions about Sanad's operations and growth.

US-06 Founder and Director | Internal Executive

As the Founder and Director, I want to review subscription and invoice records, so that I can oversee billing and see which payments are outstanding.

US-07 Founder and Director | Internal Executive

As the Founder and Director, I want to review caregiver working hours, annual background-check renewal dates, and payroll records based on completed visits, so that I can oversee staff records and pay caregivers accurately.

US-08 Elderly Member | External Operational

As an Elderly Member, I want to respond to the smart speaker when it asks whether everything is fine, so that a false alarm is cleared quickly and my family is only called when I do not answer.

US-09 Elderly Member | External Operational

As an Elderly Member, I want to pause monitoring for up to 24 hours or enable Away Mode, so that I keep my privacy and avoid unnecessary alerts when I want to be unmonitored or am travelling.

US-10 Elderly Member | External Operational

As an Elderly Member, I want to receive spoken reminders for my medication, scheduled caregiver visits, and wearing my fall detector, so that I take my medication on time, am ready for visits, and stay protected.

US-11 Family Contact | External Operational

As a Family Contact, I want to view my family member's daily wellbeing summary in my local time zone, so that I can stay informed even when I live in another city or country.

US-12 Family Contact | External Operational

As a Family Contact, I want to send voice messages to my family member's smart speaker, so that I can reassure them and stay in touch.

US-13 Family Contact | External Operational

As a Family Contact, I want to receive a phone call if my family member does not answer a safety check-in within five minutes, so that I can respond quickly when they may need help.

US-14 Family Contact | External Operational

As a Family Contact, I want to book a care visit for my family member, so that they receive help with shopping, cleaning, or companionship.

US-15 Family Contact | External Operational

As a Family Contact, I want to rate a completed visit, so that Sanad can see how well the caregiver served my family member.

US-16 Family Contact | External Operational

As a Family Contact, I want to subscribe to a care plan for my family member, so that they are covered by Sanad's monitoring and care services.

US-17 Neighbor Volunteer | External Operational

As a Neighbor Volunteer, I want to respond to non-urgent alerts in my district, so that members receive quick help from someone nearby without an emergency response.

US-18 Technology Partner | External Executive

As the Technology Partner, I want to monitor the home hub's battery status, check whether sensor events are being delivered, and review the fall detector's reported wear status, so that I can identify device or connectivity issues that may affect the reliability of the service.

US-19 Emergency Services (123) | External Executive

As an Emergency Services (123) dispatcher, I want to receive an emergency notification within two minutes of a confirmed hard fall, including the member's recorded home address and the dispatch details, so that emergency responders can be sent to the correct location without delay.

File 2: Sanad Acceptance Criteria

US 01 – Care Coordinator

AC 01.1 | Save routine information

Given: a member profile exists

When: the Care Coordinator records the member's usual wake up time, one or more meal times, and the medication schedule (medicine name, reminder time, instructions)

Then: the routine information is stored in that member's profile.

AC 01.2 | Update routine information

Given: a member has a saved routine profile

When: the Care Coordinator changes one or more routine values

Then: the updated values are saved against the same member profile.

AC 01.3 | Reject an incomplete medication entry

Given: the Care Coordinator is entering a medication schedule

When: a medication entry is submitted without a medicine name or a reminder time

Then: the entry is not saved and the missing item is indicated.

US 02 – Care Coordinator

AC 02.1 | Record informed consent before monitoring

Given: monitoring has not started for a member

When: the member provides informed consent during a video call with the Care Coordinator

Then: a Consent record is created with its consent date, status Active, review date (six months after the consent date), expiry date, and video call URI.

AC 02.2 | Require valid consent

Given: a member has no consent record, or the latest consent is Pending, Expired, or Withdrawn

When: monitoring is about to begin

Then: monitoring does not begin until valid informed consent is recorded.

AC 02.3 | Six month review

Given: a member has an active consent record with a review date

When: the six month review becomes due and the review is completed in a video call

Then: a new Consent record is added to the member's consent history with the new review and expiry dates

and status Active, and the previous record is kept unchanged.

AC 02.4 | Consent expires without review

Given: a member has an active consent record whose review was not completed

When: the expiry date is reached

Then: the consent status is set to Expired and monitoring does not continue until a new consent is recorded.

AC 02.5 | Consent withdrawn

Given: a member has an active consent record

When: the member withdraws consent

Then: a Consent record with status Withdrawn is added to the consent history and monitoring stops.

US 03 – Caregiver Team Lead

AC 03.1 | Match a caregiver to a requested visit

Given: a visit request includes a date, time, service type, and member

When: the system evaluates candidate caregivers

Then: matching considers the member's Cairo district, the required care skill, the member's preferred languages, and the caregiver's availability status being Available.

AC 03.2 | Enforce the ten hour daily limit

Given: a caregiver has existing working hours for the day

When: a proposed visit assignment would cause the caregiver to exceed ten working hours that day

Then: the system blocks that assignment and a different schedule or caregiver is required.

AC 03.3 | Assign a suitable caregiver

Given: a candidate caregiver satisfies the matching criteria and the daily work hour limit

When: the Caregiver Team Lead confirms the assignment

Then: the visit is assigned to that caregiver and the Visit record reflects the assignment.

AC 03.4 | No suitable caregiver available

Given: a visit request exists

When: no caregiver satisfies the matching criteria and the daily work hour limit

Then: the visit remains unassigned and the Caregiver Team Lead is informed so that a different date, time, or arrangement can be chosen.

US 04 – Field Caregiver

AC 04.1 | Validate check in and check out codes

Given: a visit is assigned and has check in and check out verification codes

When: the caregiver enters the correct check in code on arrival and the correct check out code on leaving

Then: the system records the actual check in time (visit status InProgress) and the actual check out time (visit status Completed).

AC 04.2 | Reject an invalid verification code

Given: a visit has an assigned verification code

When: the caregiver enters a code that does not match the code for that visit and action

Then: the system does not record that check in or check out as verified, and the visit status is unchanged.

AC 04.3 | Save the visit note

Given: the caregiver has completed the visit

When: the caregiver submits the concise visit note

Then: the note is saved in the Visit record.

AC 04.4 | Count verified hours

Given: a visit has a verified check in and a verified check out

When: the check out is recorded

Then: the verified duration between check in and check out is added to the caregiver's daily working hours and contributes to enforcing the ten hour daily limit.

US 05 – Founder and Director

AC 05.1 | Review false alerts

Given: alert records include outcomes

When: the Founder and Director reviews service performance

Then: the report shows the alerts recorded with the outcome FalseAlarm and their count.

AC 05.2 | Review average alert response time

Given: alerts have alert timestamps and resolution times

When: the Founder and Director reviews response performance

Then: the system provides the average response time derived from those timestamps; alerts without a resolution time are excluded from the average.

AC 05.3 | Review escalation outcomes

Given: alerts have recorded outcomes and escalation levels

When: the Founder and Director reviews escalation performance

Then: the report distinguishes the outcomes FalseAlarm, FamilyResolved, CaregiverHandled, and EmergencyDispatched, and the escalation levels 0 (Speaker), 1 (Family Called), 2 (Caregiver/Volunteer

Dispatched), and 3 (Emergency 123).

AC 05.4 | Review caregiver feedback

Given: completed visits have family ratings

When: the Founder and Director reviews caregiver feedback

Then: the recorded family ratings (1–5) are available for each caregiver; visits without a rating are excluded.

AC 05.5 | Review wellbeing trends

Given: a member has Monthly wellbeing reports

When: the Founder and Director reviews the monthly wellbeing trend

Then: the report presents the stored activity trend for the three month period and the related wellbeing

indicators.

AC 05.6 | No data for the selected period

Given: no alerts or visits were recorded in the selected period

When: the Founder and Director reviews service performance for that period

Then: the report shows zero alerts and no ratings instead of an error.

US 06 – Founder and Director

AC 06.1 | Review subscription details

Given: a family contact has subscribed for a member

When: the Founder and Director opens the subscription record

Then: the plan type, start date, monthly fee, and subscription status (Active, Paused, or Cancelled) are

available.

AC 06.2 | Review invoice details

Given: a subscription has generated invoices

When: the Founder and Director reviews its invoices

Then: each invoice displays its issue date, total amount, and payment status (Paid, Unpaid, or Overdue).

AC 06.3 | Identify outstanding payments

Given: a subscription has invoices with different payment statuses

When: the Founder and Director reviews its invoices

Then: invoices with payment status Unpaid or Overdue can be told apart from Paid invoices.

US 07 – Founder and Director

AC 07.1 | Review caregiver working hours

Given: a caregiver has visits during a pay period

When: the Founder and Director reviews the caregiver's payroll information

Then: the total verified hours from Completed visits are available for that pay period; Cancelled and NoShow

visits are not counted.

AC 07.2 | Review background check expiry

Given: a caregiver record exists

When: the Founder and Director reviews the caregiver record

Then: the annual background check expiry date is displayed.

AC 07.3 | Flag an expired background check

Given: a caregiver's background check expiry date has passed

When: the Founder and Director reviews the caregiver record

Then: the record displays the expiry date together with an indication that the date has passed.

AC 07.4 | Review payroll record

Given: a payroll record exists for a caregiver

When: the Founder and Director opens that record

Then: the pay period, total hours worked, amount paid, and payment date are displayed.

US 08 – Elderly Member

AC 08.1 | Receive an Arabic safety prompt

Given: an unusual non hard fall event (for example a door opened at night or no kitchen movement by the

member's routine cutoff time, based on the wake up time) triggers an alert

When: the system initiates a speaker check in

Then: the smart speaker asks the member in Arabic whether they are safe, using the configured voice speed

and pitch.

AC 08.2 | Confirm safety within five minutes

Given: the smart speaker has asked the member whether they are safe

When: the member confirms they are fine within five minutes

Then: the alert records speaker response as ConfirmedFine, outcome FalseAlarm, and alert status Closed.

AC 08.3 | Record no answer after five minutes

Given: the smart speaker has asked the member whether they are safe

When: no response is received within five minutes

Then: the alert records speaker response as NoAnswer, alert status Escalated, and escalation level 1, and

the system proceeds to the family contact escalation stage.

US 09 – Elderly Member

AC 09.1 | Pause monitoring for up to 24 hours

Given: monitoring is active

When: the member chooses to pause monitoring

Then: monitoring status is set to Paused and a pause expiry time is recorded no later than 24 hours after

the pause.

AC 09.2 | Limit a pause to 24 hours

Given: monitoring is active

When: the member asks to pause monitoring for longer than 24 hours

Then: the pause is limited to 24 hours and the member is told the limit (see Assumption A4).

AC 09.3 | Resume monitoring after a pause

Given: monitoring status is Paused

When: the pause expiry time is reached

Then: monitoring status returns to Active.

AC 09.4 | Enable Away Mode

Given: the member is travelling and enables Away Mode

When: routine empty home activity would otherwise trigger an alert

Then: routine empty home alerts are suppressed while Away Mode is enabled; hard fall alerts are not

suppressed.

AC 09.5 | Disable Away Mode

Given: Away Mode is enabled

When: the member disables Away Mode

Then: empty home alerts are evaluated again from that moment.

US 10 – Elderly Member

AC 10.1 | Receive a medication reminder

Given: a member has a saved medication schedule

When: a medication reminder time is reached

Then: the smart speaker plays the reminder with the medicine name and instructions.

AC 10.2 | Receive advance visit reminder

Given: a caregiver visit is scheduled

When: the 24 hour advance notice point is reached

Then: the smart speaker plays the visit reminder and the Visit record shows advance notice sent as true.

AC 10.3 | Receive pendant wear reminder

Given: the fall detector reports a wear status of Unworn or Charging during daytime hours (see Assumption

A1)

When: the system processes that status

Then: the smart speaker plays an Arabic reminder asking the member to wear the fall detector.

US 11 – Family Contact

AC 11.1 | View daily wellbeing summary

Given: a family contact is linked to a member and a Daily Wellbeing Report exists

When: the family contact opens the daily summary

Then: the dashboard displays safety status (Green or Red), sleep status, meal status, and visit completion status.

AC 11.2 | Apply the family contact's local time zone

Given: the family contact has a time zone recorded

When: the daily summary is displayed

Then: its date and time presentation is adjusted to that recorded local time zone.

AC 11.3 | Daily summary not yet available

Given: a family contact is linked to a member and no Daily Wellbeing Report exists for the day

When: the family contact opens the daily summary

Then: the dashboard states that the summary is not yet available.

AC 11.4 | Restrict summaries to linked members

Given: a family contact is not linked to a member

When: the family contact tries to view that member's summary

Then: the system does not show the summary.

US 12 – Family Contact

AC 12.1 | Send a voice message

Given: a family contact is linked to a member

When: the family contact sends a voice message to that member

Then: an Audio Message record is created with the audio URI, sent timestamp, and delivery status Queued, and the message is queued for playback on the member's smart speaker.

AC 12.2 | Show voice message delivery status

Given: a voice message has been sent

When: the delivery status changes

Then: the status is updated to Sent, Delivered, Played, or Failed and is visible to the family contact.

AC 12.3 | Restrict messages to linked members

Given: a family contact is not linked to a member

When: the family contact tries to send a voice message to that member

Then: the system does not accept the message.

US 13 – Family Contact

AC 13.1 | Call family contacts in priority order

Given: the member has not confirmed safety within five minutes

When: the family escalation stage begins

Then: the system calls the linked family contacts according to their recorded call order and records for each

step who was reached, when, and whether they answered.

AC 13.2 | Family contact answers the call

Given: a family contact is called during escalation

When: the family contact answers and confirms that the member is safe or that help has been arranged

Then: the escalation step is recorded as answered, further escalation stops, and the alert outcome is

recorded as FamilyResolved.

AC 13.3 | Continue escalation when family is unreachable

Given: family contacts have been called in priority order

When: family remains unreachable for ten minutes

Then: the system dispatches the nearest available caregiver for the urgent alert and sets escalation level to

2.

AC 13.4 | Record caregiver arrival

Given: a caregiver has been dispatched to an alert

When: the caregiver arrives at the member's home

Then: the arrival time is recorded so that it can be compared with the 45 minute Cairo target.

US 14 – Family Contact

AC 14.1 | Book a visit

Given: a family contact is linked to a member

When: the family contact submits a visit request with a date, start and end time, and service type (Shopping,

Cleaning, or Companionship)

Then: a Visit record is created for that member with visit status Scheduled and is available to the Caregiver

Team Lead for assignment (see Assumption A7).

AC 14.2 | Reject an incomplete visit request

Given: a family contact is booking a visit

When: the request is missing the date, start time, end time, or service type

Then: the visit is not created and the missing item is indicated.

AC 14.3 | Restrict booking to linked members

Given: a family contact is not linked to a member

When: the family contact tries to book a visit for that member

Then: the system does not create the visit.

US 15 – Family Contact

AC 15.1 | Rate a completed visit

Given: a visit has visit status Completed

When: the family contact submits a rating from 1 to 5

Then: the family rating is saved in the Visit record.

AC 15.2 | Reject a rating for an uncompleted visit

Given: a visit is not Completed

When: the family contact tries to submit a rating

Then: the rating is not saved.

AC 15.3 | Reject a rating outside 1 to 5

Given: a visit has visit status Completed

When: the family contact submits a rating below 1 or above 5

Then: the rating is not saved and the allowed range is shown.

US 16 – Family Contact

AC 16.1 | Subscribe for a member

Given: a family contact is linked to a member who has no subscription

When: the family contact chooses a plan

Then: a Subscription record is created with its plan type, start date, monthly fee, and status Active, paid for

by that family contact.

AC 16.2 | Generate monthly invoices

Given: a subscription is Active

When: a new month of the subscription begins

Then: an Invoice record is created with its issue date, total amount, and payment status Unpaid (see

Assumption A6).

AC 16.3 | One subscription per member [Confirm with team]

Given: a member is already covered by a subscription

When: another subscription is requested for that member

Then: the new subscription is not created.

US 17 – Neighbor Volunteer

AC 17.1 | Notify suitable volunteers

Given: a non urgent alert has occurred for a member

When: the system looks for volunteers to respond

Then: only Neighbor Volunteers with verification status Verified in the member's district are notified (see

Assumption A8).

AC 17.2 | Respond to an alert

Given: a Verified volunteer has been notified of a non urgent alert

When: the volunteer accepts and responds

Then: the alert records which volunteer responded and the escalation level 2.

AC 17.3 | Do not use volunteers for urgent alerts

Given: an alert is urgent (a hard fall, or family unreachable)

When: the system chooses who responds

Then: the alert goes to emergency services or the nearest available caregiver, not to a volunteer.

US 18 – Technology Partner

AC 18.1 | Review hub operating status

Given: a Home Hub is installed for a member

When: hub status is reviewed

Then: the recorded battery status (0–100%), power status (MainsAvailable or PowerOutage), and

connection status (Online or Offline) are available.

AC 18.2 | Review sensor status and type

Given: sensors are connected to a home hub

When: sensor information is reviewed

Then: each sensor's type, room, sensor status, and wear status (N/A for non wearable sensors) are

available.

AC 18.3 | Record sensor events

Given: an installed sensor reports an event

When: the system receives the event

Then: a Sensor Event record stores the event type, event timestamp, event details, and the reporting

sensor, and the latest event time of each sensor can be reviewed.

AC 18.4 | Identify an offline hub

Given: a hub's connection status is Offline

When: the Technology Partner reviews hub status

Then: the hub is shown as Offline so that a connectivity issue can be identified.

AC 18.5 | Identify a faulty or offline sensor

Given: a sensor's sensor status is Faulty or Offline

When: the Technology Partner reviews sensor status

Then: the sensor is shown as needing attention.

AC 18.6 | Raise a low battery alert

Given: a hub battery falls below the low battery level (see Assumption A2)

When: the system checks hub status

Then: an alert of type HubLowBattery is created for the member without a sensor event, with

speaker response NotApplicable.

AC 18.7 | Deliver sensor events promptly

Given: an installed sensor generates an event

When: the event is reported through the home hub

Then: the event is received by the system within five seconds.

US 19 – Emergency Services (123)

AC 19.1 | Dispatch emergency services within two minutes

Given: a hard fall is detected for a member

When: the system processes the hard fall event

Then: it initiates the emergency call to service 123 within two minutes, sets escalation level to 3, and

records dispatch time, service number, and dispatch status.

AC 19.2 | Include address and dispatch details

Given: an emergency call to 123 is initiated

When: the notification is given

Then: it includes the member's recorded home address (street, district, city) and the dispatch details (see

Assumption A3).

AC 19.3 | Do not delay a hard fall dispatch

Given: a hard fall is detected

When: the system starts the response

Then: emergency dispatch does not wait for a speaker check in or for any family response.

AC 19.4 | Notify family in parallel

Given: a hard fall event has been detected

When: the system initiates emergency dispatch

Then: family contacts are notified at the same time; family notification does not wait for the emergency call to

finish or for a family contact to answer.

AC 19.5 | Emergency call not connected

Given: an emergency call to 123 has been initiated

When: the call is not connected

Then: dispatch status records that the call was not connected and the alert outcome is not set to EmergencyDispatched.

AC 19.6 | Record emergency outcome

Given: an emergency call to 123 has been connected

When: the dispatch status or outcome is updated

Then: the Alert record reflects the recorded emergency dispatch status. The outcome is set to EmergencyDispatched only when the dispatch is confirmed as successful.

User Story Quality Evaluation

1. Evaluation Methodology

To ensure that the elicited requirements are well-formed, feasible, and rigorous, all nineteen user stories were evaluated against the INVEST criteria:

- **Independent (I):** Can the story be developed and tested with minimal coupling to other backlog items? Dependencies that reflect natural workflow sequencing (e.g., booking prior to rating) are noted, whereas hard technical blocks that impede separate scheduling are flagged.
- **Negotiable (N):** Does the story capture the underlying stakeholder need and business constraint without prematurely prescribing user interface mechanics or implementation technology?

- **Valuable (V):** Is the business or user value clearly articulated and directly traceable to elicitation findings (e.g., stakeholder interviews, Sanad Care Charter)?
- **Estimable (E):** Is the story sufficiently unambiguous, bounded, and supported by explicit domain rules for developers to gauge the implementation effort?
- **Small (S):** Is the story appropriately scoped to represent a manageable work item, rather than aggregating multiple distinct triggers, user actions, or operational lifecycles?
- **Testable (T):** Are the acceptance criteria clear, verifiable, and comprehensive, covering expected behavior as well as alternative paths and exceptional/error conditions?

2. Master INVEST Evaluation Matrix

Quality evaluation across all nineteen user stories. Stories flagged under Small (S) reflect high-level stakeholder epics; their decomposition is detailed in Section 4.

Story ID	I	N	V	E	S	T	Overall Verdict
US-01	Pass	Pass	Pass	Pass	Pass	Pass	Pass. Cohesive data setup transaction.
US-02	Partial	Pass	Pass	Pass	Fail	Pass	Epic. Scoped at lifecycle level; expiry rule formalized.
US-03	Partial	Pass	Pass	Pass	Pass	Pass	Pass. Workload rule clarified across verified/scheduled hours.
US-04	Partial	Pass	Pass	Pass	Fail	Pass	Epic. Combines verification codes with narrative reporting.
US-05	Partial	Pass	Pass	Pass	Fail	Pass	Epic. Multi-domain dashboard; metrics and query windows bounded.
US-06	Partial	Pass	Pass	Pass	Pass	Pass	Pass. Focused financial ledger query.
US-07	Partial	Pass	Pass	Pass	Fail	Pass	Epic. Combines payroll verification with HR compliance tracking.

US-08	Partial	Pass	Pass	Pass	Pass	Pass	Pass. Interactive dialog; distress response path added.
US-09	Partial	Pass	Pass	Pass	Fail	Pass	Epic. Combines timer pause and state toggle; fall behavior defined.
US-10	Partial	Pass	Pass	Pass	Fail	Pass	Epic. Spans three discrete event triggers; daytime window bounded.
US-11	Partial	Pass	Pass	Pass	Pass	Pass	Pass. Read-only dashboard with timezone conversion.
US-12	Pass	Pass	Pass	Pass	Pass	Pass	Pass. Independent asynchronous media transmission.
US-13	Partial	Pass	Pass	Pass	Fail	Pass	Epic. Combines priority calling sequence with field dispatch.
US-14	Pass	Pass	Pass	Pass	Pass	Pass	Pass. Single booking transaction; temporal validations tightened.
US-15	Partial	Pass	Pass	Pass	Pass	Pass	Pass. Exemplar atomic story with complete edge case coverage.
US-16	Pass	Pass	Pass	Pass	Fail	Pass	Epic. Plan enrollment bundles recurring invoicing; constraints set.
US-17	Partial	Pass	Pass	Pass	Pass	Pass	Pass. Discrete response action; timeout escalation fallback added.
US-18	Pass	Pass	Pass	Pass	Fail	Pass	Epic. Combines hardware infrastructure with real-time stream SLA.
US-19	Partial	Pass	Pass	Pass	Pass	Pass	Pass. Automated emergency integration; trigger wording harmonized.

3. Category 1: Internal Operational Roles

US-01: Care Coordinator – Daily Routine Configuration

Story: As a Care Coordinator, I want to set up and update each member's daily routine, including wake-up time, meal times, and medication reminders, so that the system uses the member's normal routine when it monitors for unusual activity.

Evaluation & refinements: Satisfies all criteria. AC 01.1–01.3 cover creation, updates, and rejection of incomplete medication schedules.

US-02: Care Coordinator – Informed Consent & Six-Month Reviews

Story: As a Care Coordinator, I want to record each member's informed consent and complete the six-month consent reviews, so that monitoring only takes place with valid consent.

Evaluation & refinements: I Partial (monitoring cannot begin without valid consent). S Fail: three temporal operations (initial consent, bi-annual review, on-demand withdrawal). Fix (AC 02.1 & 02.4): review_date = consent_date + 6 calendar months; expiry_date = review_date + 30 days (grace period). Decomposition: US-02A Initial Consent Recording, US-02B Bi-annual Review & Expiry, US-02C Consent Withdrawal.

US-03: Caregiver Team Lead – Caregiver Visit Assignment

Story: As a Caregiver Team Lead, I want to assign caregivers to requested visits based on district, skills, language, availability, and the ten-hour daily working-hours limit, so that members receive suitable care and no caregiver is overworked.

Evaluation & refinements: I Partial (downstream of visit booking). Clarification (AC 03.2): Daily Allocated Hours = sum(Verified Completed Hours) + sum(Scheduled Pending Hours). If Daily Allocated Hours + Proposed Duration > 10.0 hours, assignment is blocked. AC 03.1–03.4 cover matches, hour ceiling, confirmation, and unassigned fallback.

US-04: Field Caregiver – Visit Verification & Reporting

Story: As a Field Caregiver, I want to confirm my arrival and departure with check-in and check-out codes and submit a short visit note, so that completed visits are properly verified and recorded.

Evaluation & refinements: I Partial (needs an assigned visit). S Fail: conflates presence verification with clinical visit notes. Decomposition: US-04A Check-In/Check-Out Code Verification, US-04B Post-Visit Note Submission.

3. Category 2: Internal Executive Roles

US-05: Founder and Director – Service Quality & Operational Analytics

Story: As the Founder and Director, I want to review service information such as the number of false alerts, average alert response time, how alerts were resolved, caregiver family ratings, and members' activity trends, so that I can assess service quality and make informed decisions about Sanad's operations and growth.

Evaluation & refinements: S Fail: bundles five reporting sub-domains. Fix (AC 05.5): reporting window bounded to monthly/quarterly; Delta Activity = ((Sensor Counts in Window − Baseline Counts) / Baseline Counts) × 100%. Decomposition: US-05A Incident Response & Escalation Analytics, US-05B Caregiver Quality & Ratings, US-05C Member Longitudinal Wellbeing Trends.

US-06: Founder and Director – Subscription & Invoicing Oversight

Story: As the Founder and Director, I want to review subscription and invoice records, so that I can oversee billing and see which payments are outstanding.

Evaluation & refinements: Satisfies all criteria. AC 06.1–06.3 verify record viewing, status categorization, and outstanding balance highlighting.

US-07: Founder and Director – Staff Hours, Payroll & Compliance

Story: As the Founder and Director, I want to review caregiver working hours, annual background-check renewal dates, and payroll records based on completed visits, so that I can oversee staff records and pay caregivers accurately.

Evaluation & refinements: S Fail: combines bi-weekly payroll reconciliation with annual background-check compliance. Decomposition: US-07A Payroll Reconciliation from Verified Hours, US-07B Background Check Compliance & Renewal Tracking.

3. Category 3: External Operational Roles

US-08: Elderly Member – Smart Speaker Safety Check-in

Story: As an Elderly Member, I want to respond to the smart speaker when it asks whether everything is fine, so that a false alarm is cleared quickly and my family is only called when I do not answer or need help.

Evaluation & refinements: Wording refined to capture distress. New AC 08.4: Given the speaker has prompted the member in Arabic, When the member responds indicating distress ("لا، أحتاج مساعدة" / "No, I need help"), Then the alert records the response, sets the

status, and initiates immediate family telephony escalation without waiting for the 5-minute timeout.

US-09: Elderly Member – Privacy Pause & Away Mode

Story: As an Elderly Member, I want to pause monitoring for up to 24 hours or enable Away Mode, so that I keep my privacy and avoid unnecessary alerts when I want to be unmonitored or am travelling.

Evaluation & refinements: S Fail: Privacy Pause (self-reverting timer, max 24 h) vs Away Mode (state latch until deactivated). AC 09.1: hard-fall detection stays active during a pause; only routine anomaly checks are suspended. AC 09.4: routine empty-home alerts = door sensor triggers, prolonged bed absence, kitchen inactivity timers. New AC 09.6: pause expiring during an open alert reverts without duplicate retrospective alerts. Decomposition: US-09A Temporary Privacy Pause (24 h auto-expiry), US-09B Extended Away Mode Toggle.

US-10: Elderly Member – Multimodal Spoken Reminders

Story: As an Elderly Member, I want to receive spoken reminders for my medication, scheduled caregiver visits, and wearing my fall detector, so that I take my medication on time, am ready for visits, and stay protected.

Evaluation & refinements: S Fail: three event drivers (clock schedule, 24-hour advance trigger, telemetry change). AC 10.3: daytime = wake-up time to 22:00 local (default 08:00–22:00). Medication and visit reminders are suppressed during Away Mode. New AC 10.4: visit cancelled before the 24-hour trigger → reminder dequeued, no stale audio. Decomposition: US-10A Medication, US-10B 24-Hour Visit Reminders, US-10C Daytime Pendant Wear Reminders.

US-11: Family Contact – Daily Wellbeing Summary

Story: As a Family Contact, I want to view my family member's daily wellbeing summary in my local time zone, so that I can stay informed even when I live in another city or country.

Evaluation & refinements: Satisfies all criteria. AC 11.1–11.4: rendering, timezone localization, pending report states, authorization limited to linked members.

US-12: Family Contact – Voice Messaging

Story: As a Family Contact, I want to send voice messages to my family member's smart speaker, so that I can reassure them and stay in touch.

Evaluation & refinements: Satisfies all criteria. AC 12.1–12.3: audio queuing, delivery status transitions, link verification.

US-13: Family Contact – Safety Escalation Telephony

Story: As a Family Contact, I want to receive a phone call if my family member does not answer a safety check-in within five minutes, so that I can respond quickly when they may need help.

Evaluation & refinements: S Fail: sequential family calling plus field dispatch. AC 13.3: emergency field caregiver dispatch after family unreachability = Escalation Level 2B (Field Caregiver). Decomposition: US-13A Sequential Family Telephony Escalation, US-13B Caregiver Urgent Fallback Dispatch.

US-14: Family Contact – Care Visit Booking

Story: As a Family Contact, I want to book a care visit for my family member, so that they receive help with shopping, cleaning, or companionship.

Evaluation & refinements: Fix (AC 14.2): validation rule start_time > current_time AND end_time > start_time; past or zero/negative-duration bookings are rejected with descriptive feedback.

US-15: Family Contact – Rate Completed Visit

Story: As a Family Contact, I want to rate a completed visit, so that Sanad can see how well the caregiver served my family member.

Evaluation & refinements: Satisfies all criteria, no changes. AC 15.1 valid path; AC 15.2 prerequisite state (completed visit); AC 15.3 boundary $1 \leq \text{rating} \leq 5$.

US-16: Family Contact – Care Plan Subscription

Story: As a Family Contact, I want to subscribe to a care plan for my family member, so that they are covered by Sanad's monitoring and care services.

Evaluation & refinements: S Fail: subscription agreement plus automated monthly invoicing. New AC 16.3: Given a member is covered by an Active subscription, When another subscription request is submitted, Then it is rejected, no duplicate is created, and the user is told an active plan already covers the member. Decomposition: US-16A Plan Selection & Activation, US-16B Automated Monthly Invoice Engine.

US-17: Neighbor Volunteer – Non-Urgent Alert Response

Story: As a Neighbor Volunteer, I want to respond to non-urgent alerts in my district, so that members receive quick help from someone nearby without an emergency response.

Evaluation & refinements: AC 17.2: volunteer claim = Escalation Level 2A (Community Volunteer Response). New AC 17.4: if no verified volunteer accepts within 15 minutes, the alert is re-escalated to the primary Family Contact and the nearest available Field Caregiver (Level 2B) is notified.

3. Category 4: External Executive Roles

US-18: Technology Partner – Hub & Sensor Telemetry Monitoring

Story: As the Technology Partner, I want to monitor the home hub's battery status, check whether sensor events are being delivered, and review the fall detector's reported wear status, so that I can identify device or connectivity issues that may affect the reliability of the service.

Evaluation & refinements: S Fail: hub status, sensor telemetry, and pipeline latency. Decomposition: US-18A Home Hub Power, Battery & Connectivity Monitoring, US-18B Sensor Diagnostics & Telemetry Latency SLA (5 seconds).

US-19: Emergency Services (123) – Automated Hard Fall Notification

Story: As an Emergency Services (123) dispatcher, I want to receive an emergency notification within two minutes of a detected hard fall, including the member's recorded home address and the dispatch details, so that emergency responders can be sent to the correct location without delay.

Evaluation & refinements: Terminology aligned: story now says "detected hard fall" (not "confirmed"), matching AC 19.1; AC 19.6 records dispatch confirmation on connection.

4. Story Granularity and Decomposition

Nine of nineteen stories (US-02, 04, 05, 07, 09, 10, 13, 16, 18) fail the Small criterion because they were written at the stakeholder-goal/epic level. They were kept at goal level for three reasons: (1) elicitation traceability to stakeholder statements and the Sanad Care Charter; (2) domain entity cohesion (e.g., consent and subscription lifecycles); (3) decomposition through the Event Model in Part 2 (e.g., US-10 → E26 medication time, E28 24-hour visit notice, E41 detector unworn).

Original Epic	Atomic Stories	Core Responsibility
US-02	US-02A, 02B, 02C	Record initial consent and video-call metadata; run 6-month reviews and enforce expiry; process withdrawal and stop monitoring.
US-04	US-04A, 04B	Verify arrival/departure with dual-OTP codes; store post-visit notes.
US-05	US-05A, 05B, 05C	Alert response times, false alarms, resolution paths; caregiver and rating metrics; longitudinal activity variance vs baseline.
US-07	US-07A, 07B	Reconcile verified hours into payroll; audit background-check validity and renewals.
US-09	US-09A, 09B	Temporary pause with 24-hour auto-expiry; Away Mode toggle.
US-10	US-10A, 10B, 10C	Medication announcements; 24-hour advance visit notices; daytime detector-unworn reminders.
US-13	US-13A, 13B	Sequential phone dialing of family; urgent caregiver dispatch after 10-minute unreachability.
US-16	US-16A, 16B	Plan selection and act ive-plan validation; automated monthly invoices and payment status.
US-18	US-18A, 18B	Hub power, battery, connectivity; sensor health and 5-second delivery SLA.

Event Analysis and Event-Use Case Table

1. Perfect Technology Assumption

All identified events and use cases were reviewed to confirm each is a genuine business requirement. Acceptance criteria that only handle technology failure or performance were not turned into events (table below). Technical and security functions (login/logout, change password, backup/restore database) are also excluded. E25 (device status review) is kept only as a read-only query because it is a stated stakeholder need (US-18).

Excluded item	**Source**	**Reason**
Hub or sensor offline or faulty	US-18; AC 18.4, 18.5	Handles a device fault, not a business activity
Hub low-battery alert and power outage	US-18; AC 18.6	Handles a power and hardware problem
Sensor events delivered within 5 seconds	US-18; AC 18.7	Performance requirement, not a business event
Emergency call to 123 not connected	US-19; AC 19.5	Handles a communication failure
Voice-message delivery status (Failed, Delivered, etc.)	US-12; AC 12.2	Handles a delivery failure

2–5. Event–Use Case Table (External, Temporal and State events)

Events E01–E25 are external, E26–E38 temporal, E39–E42 state events.

Event ID	Event Description	Event Type	Triggering Actor / Condition	Corresponding Use Case
E01	Routine information is set or changed	External	Care Coordinator	Manage routine profile
E02	Member gives informed consent	External	Care Coordinator (with Elderly Member)	Record informed consent
E03	Six-month consent review is completed	External	Care Coordinator	Complete consent review
E04	Member withdraws consent	External	Elderly Member	Withdraw consent
E05	Care visit is requested	External	Family Contact; Caregiver Team Lead	Book care visit
E06	Caregiver is assigned to a visit	External	Caregiver Team Lead	Assign caregiver to visit
E07	Caregiver arrives and checks in	External	Field Caregiver	Check in to visit
E08	Caregiver leaves and checks out	External	Field Caregiver	Check out of visit
E09	Caregiver submits visit note	External	Field Caregiver	Submit visit note

E10	Family rates a completed visit	External	Family Contact	Rate completed visit
E11	Family subscribes to a care plan	External	Family Contact	Subscribe to care plan
E12	Family opens daily summary	External	Family Contact	View daily wellbeing summary
E13	Family sends a voice message	External	Family Contact	Send voice message
E14	Member answers the speaker check-in	External	Elderly Member	Confirm safety
E15	Member pauses monitoring	External	Elderly Member	Pause monitoring
E16	Member enables Away Mode	External	Elderly Member	Enable Away Mode
E17	Member disables Away Mode	External	Elderly Member	Disable Away Mode
E18	Family contact answers escalation call	External	Family Contact	Record family response
E19	Dispatched caregiver arrives at member's home	External	Field Caregiver	Record urgent-visit arrival
E20	Volunteer responds to an alert	External	Neighbor Volunteer	Respond to non-urgent alert

E21	Emergency Services confirm dispatch	External	Emergency Services (123)	Record emergency dispatch outcome
E22	Director requests service performance report	External	Founder and Director	Review service performance
E23	Director reviews subscriptions and invoices	External	Founder and Director	Review subscriptions and invoices
E24	Director reviews caregiver records and payroll	External	Founder and Director	Review caregiver records and payroll
E25	Technology Partner requests device status (query only under PTA)	External	Technology Partner	Review device and sensor status
E26	Medication reminder time reached	Temporal	Daily, at each reminder_time in the member's medication schedule.	Play medication reminder
E27	Morning routine cutoff passes	Temporal	Daily, at the member's morning cutoff (derived from wake-up time; exact cutoff, e.g. 11:30 or 12:00, to be confirmed with the Care Coordinator). Checked against kitchen-sensor state.	Check morning inactivity
E28	24-hour advance-notice point reached	Temporal	Once per scheduled visit, at scheduled_time minus 24 hours.	Send advance visit notice

E29	End of day (daily summary time)	Temporal	Daily, once per member at a fixed end-of-day time (to be confirmed). Display adjusted to the family contact's time zone.	Generate daily wellbeing report
E30	End of month (monthly report time)	Temporal	Monthly, once per member; report presents a three-month activity trend.	Generate monthly wellbeing report
E31	Six-month consent review date reached	Temporal	Once per consent record, at review_date (consent_date plus 6 months).	Flag consent review as due
E32	Consent expiry date reached without review	Temporal	Once per consent record, at expiry_date, if the review was not completed.	Expire consent and stop monitoring
E33	Monitoring pause expires	Temporal	Once per pause, at pause_expiry_time (no later than 24 hours after the pause began).	Resume monitoring after pause
E34	Subscription month begins	Temporal	Monthly, on each monthly anniversary of an Active subscription's start_date.	Generate monthly invoice
E35	Pay period ends	Temporal	End of each pay period (length to be confirmed), per caregiver.	Calculate caregiver payroll
E36	Background-check expiry date reached	Temporal	Annually, per caregiver, at bg_check_expiry_date.	Flag expired background check
E37	Speaker check-in response window elapses	Temporal	5 minutes after the Arabic safety prompt, if no confirmation was received.	Escalate alert to family contacts
E38	Family-unreachable window elapses	Temporal	10 minutes of family contacts remaining unreachable (report does not state whether counted	Dispatch nearest available caregiver

			from the alert or the first call; AC 13.3 implies from the family calls).	
E39	Hard fall detected	State	Fall status changes from normal to hard fall detected while monitoring is active (not suppressed by Away Mode).	Dispatch emergency services (123) and notify family
E40	Unusual non-fall activity detected	State	Sensor activity changes from matching the member's routine to deviating from it (e.g. door opened at night), while monitoring is active and Away Mode is off.	Start speaker safety check-in
E41	Fall detector unworn or charging in daytime	State	Wear status changes from Worn to Unworn or Charging during daytime hours.	Send fall detector wear reminder
E42	Non-urgent alert raised	State	A new alert is created with a non-urgent type, and verified volunteers exist in the member's district.	Notify neighbor volunteers